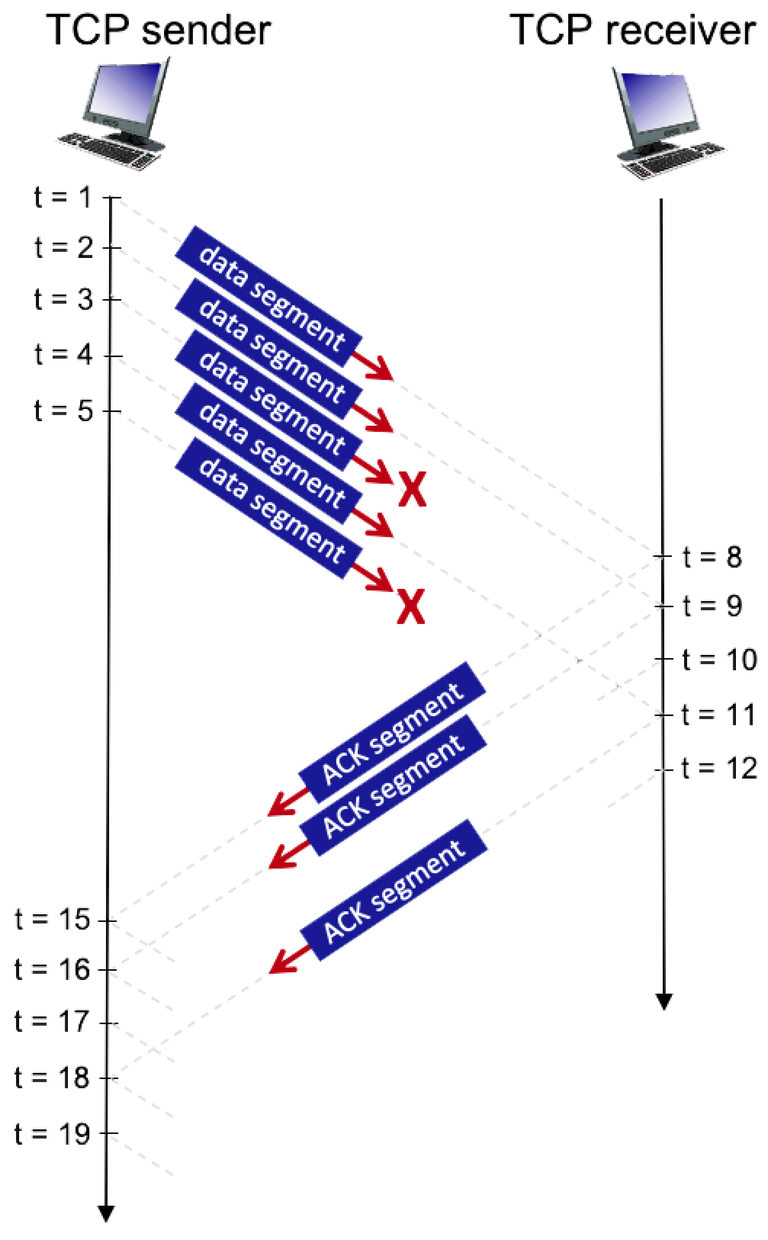
INTERACTIVE END-OF-CHAPTER EXERCISES

TCP RETRANSMISSIONS (RELIABLE DATA TRANSMISSION WITH ACK LOSS)

Consider the figure below in which a TCP sender and receiver communicate over a connection in which the segments can be lost. The TCP sender wants to send a total of 10 segments to the receiver and sends an initial window of 5 segments at t = 1, 2, 3, 4, and 5, respectively. Suppose the initial value of the sequence number is 21 and every segment sent to the receiver each contains 256 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at t = 8, and an ACK for this segment arrives at t = 15. As shown in the figure, 2 of the 5 segments is lost between the sender and the receiver, but *none* of the ACKs are lost. Assume there are no timeouts and any out of order segments received are thrown out.



QUESTION LIST

1. What is the sequence number of the segment sent at t=1?

2. What is the sequence number of the segment sent at t=2?

3. What is the sequence number of the segment sent at t=3?

4. What is the sequence number of the segment sent at t=4?

5. What is the sequence number of the segment sent at t=5?

6. What is the value of the ACK sent at t=8? (If segment lost, write 'x')

7. What is the value of the ACK sent at t=9? (If segment lost, write 'x')

8. What is the value of the ACK sent at t=10? (If segment lost, write 'x')

9. What is the value of the ACK sent at t=11? (If segment lost, write 'x')

10. What is the value of the ACK sent at t=12? (If segment lost, write 'x')

11. What is the sequence number of the segment sent at t = 15? (If ACK never arrives, write 'x')

12. What is the sequence number of the segment sent at t = 16? (If ACK never arrives, write 'x')

13. What is the sequence number of the segment sent at t = 17? (If ACK never arrives, write 'x')

14. What is the sequence number of the segment sent at t = 18? (If ACK never arrives, write 'x')

15. What is the sequence number of the segment sent at t = 19? (If ACK never arrives, write 'x')

SOLUTION

1. The sequence number of the first segment is the starting sequence number, which is 21.

2. The sequence number of the second segment is = 21 + 256 = 277.

3. The sequence number of the third segment is = 277 + 256 = 533.

4. The sequence number of the fourth segment is = 533 + 256 = 789.

5. The sequence number of the fifth segment is = 789 + 256 = 1045.

6. The ACK value is the sequence number of the next expected segment, which is 277.

7. The ACK value is the sequence number of the next expected segment, which is 533.

8. Since the segment was lost, the ACK is never sent, so the answer is x.

9. The ACK value is the sequence number of the next expected segment, which is 533.

10. Since the segment was lost, the ACK is never sent, so the answer is x.

11. The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 1301.

12. The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 1557.

13. Since there was nothing to send (as we're waiting for ACKs), the answer is x.

14. Since there was nothing to send (as we're waiting for ACKs), the answer is x.

15. Since there was nothing to send (as we're waiting for ACKs), the answer is x.

QUESTION 1 OF 15

What is the sequence number of the segment sent at t=1?

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

Copyright © 2010-2025 J.F. Kurose, K.W. Ross
Comments welcome and appreciated: kurose@cs.umass.edu